

Aedan Cook-Shen

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WORK & LEADERSHIP EXPERIENCE

Professor Maziar Ghazinejad's Research Lab

San Diego, California

Undergraduate Research Assistant

March 2026 – Present

- Led a 4-person computer vision research project, improving deep learning model segmentation performance for nanocrystalline alumina characterization by 2.4% through a novel physics-inspired data augmentation approach.
- Conducted 170 deep neural network trainings using Python, TensorFlow/Keras, OpenCV, and NumPy, applying Bayesian optimization and controlled experiments to evaluate segmentation performance and robustness.
- Standardized machine learning experimentation and data workflows by containerizing the research pipeline with Docker, ensuring consistent dependencies across local development and remote Linux GPU training.
- Presented first-author research at SPIE Optics + Photonics 2026 on automated segmentation and grain detection in resolution-limited nanocrystalline alumina SEM images.

UCSD Information Technology Services (ITS) Service Desk

San Diego, California

Service Desk Technician

September 2026 – Present

- Troubleshoot user-reported IT issues through support calls and service tickets, including Active Directory-related requests and other technical support needs.

UCSD Outback Adventures – Challenge Course

San Diego, California

Team Facilitator

June 2025 – Present

- Delivered experiential learning programs focused on team communication and collaboration for 700+ participants across 18 recurring client groups, including university teams, MBA cohorts, and youth programs.
- Led challenge course and high-ropes activities for groups of up to 104 participants, coordinating across facilitators, communicating safety-critical procedures, and adapting execution in real time to changing conditions and participant needs.
- Managed course setup and equipment preparation for recurring programs, verifying activity configurations and safety systems before participant use.

EDUCATION

University of California San Diego

San Diego, California

B.S. in Data Science

Expected June 2028

- **GPA:** 3.96 / 4.0
- **Relevant Coursework:** Data Structures and Algorithms, Principles of Data Science, Computational Physics: Probabilistic Models & Simulations, Data Science Tools & Workflows, Linear Algebra

SKILLS & INTERESTS

Programming: Python, Java, C#, Javascript

Development Tools: Git, Linux, Bash, Docker, uv, pytest

Libraries & Frameworks: PyTorch, TensorFlow/Keras, OpenCV, NumPy

Cloud: AWS (Certified Cloud Practitioner), System Troubleshooting, Identity & Access Management